

Problem 1. Let $\varphi : G_1 \rightarrow G_2$ be a surjective group homomorphism.

- (a) Suppose that $G_1 = \mathbb{Z}/39\mathbb{Z}$ and $|G_2| = 13$. Determine with proof the kernel of φ .

Proof. Since φ is a group homomorphism, we know by the first isomorphism theorem that

$$\begin{aligned} G_1/\ker(\varphi) \cong \varphi(G_1) = G_2 &\implies |G_1/\ker(\varphi)| = |G_2| \\ &\implies \frac{|G_1|}{|\ker(\varphi)|} = |G_2| \\ &\implies |\ker(\varphi)| = \frac{|G_1|}{|G_2|}. \end{aligned}$$

Since $|G_1| = 39$ and $|G_2| = 13$, then $|\ker(\varphi)| = 3$. This tells us that $\ker(\varphi) = \langle \overline{13} \rangle$. ■

- (b) Suppose that $G_1 = D_8$ and $G_2 = \mathbb{Z}/2\mathbb{Z}$. Determine the possible kernels of φ .

Proof. Again, since φ is a group homomorphism, the first isomorphism theorem tells us that

$$\frac{|G_1|}{|\ker(\varphi)|} = 2 \implies |\ker(\varphi)| = \frac{8}{2} = 4.$$

This tells us that we need to look for all possible subgroups of D_8 of order 4. From the lattice of D_8 , we can see that the possible subgroups that $\ker(\varphi)$ can be are $\langle r \rangle$, $\langle sr^2 \rangle$, and $\langle rs, r^2 \rangle$. ■

- (c) Suppose that $|G_1|$ and $|G_2|$ are relatively prime. Prove that $G_2 = \{1_{G_2}\}$.

Proof. Using the first isomorphism theorem and the fact that φ is a surjective group homomorphism, it follows that

$$\frac{|G_1|}{|\ker(\varphi)|} = |G_2| \implies |G_1| = |G_2||\ker(\varphi)|.$$

Since $(|G_1|, |G_2|) = 1$, it follows that $|G_2| = 1$ and so $G_2 = \{1_{G_2}\}$. ■

Problem 2. Let G be a group with maximal subgroup H . Recall that H is maximal means that $H \neq G$ and the only subgroups which contain H are H and itself.

- (a) Prove that if $|G : H| = p$ where p is prime, then H is a maximal subgroup of G .

Proof. Our goal is to show that H is a maximal subgroup of G ; that is, we need to show that for all $K \leq G$ such that $H \leq K \leq G$, either $H = K$ and $K = G$. To this end, let $K \leq G$ be such that $H \leq K \leq G$. By the Tower Laws, it follows that

$$\begin{aligned} |G : H| &= |G : K||K : H| \\ &\implies p = |G : K||K : H|. \end{aligned}$$

Since p is prime, either $|G : K| = 1$ or $|K : H| = 1$ since the above shows that either $|G : K|$ or $|K : H|$ divides p . In either case, $G = K$ or $K = H$. This tells us that H must be maximal. ■

- (b) Prove that if $H \trianglelefteq G$, then G/H is cyclic. Deduce that $|G : H|$ is prime.

Proof. Suppose that $H \trianglelefteq G$. First, notice that since $H \trianglelefteq G$, the group structure of G/H is well-defined. Since H is maximal, it follows that $H \neq G$. Thus, take an element $g \notin H$. Since $g \notin H$, it follows that $gH \neq 1H$. Consider the cyclic subgroup of G/H generated by gH denoted by $\langle gH \rangle$. Since H is maximal, we know that either $\langle gH \rangle = G/H$ or $\langle gH \rangle = \bar{1}$. Since

$gH \neq 1H$, we know that $\langle gH \rangle = G/H$ and so G/H must be cyclic. Consider the index of H in G denoted by $|G : H|$.

We need to show that $|G : H|$ is prime. Suppose for contradiction that $|G : H|$ is composite. Then $|G : H| = pq$ where p and q are primes. Since q divides $|G : H|$, it follows that there exists a subgroup K/H of G/H of order q . By the Fourth Isomorphism Theorem, we see that $K \leq G$. Since H is maximal, $K = H$. That is, $K/H = \bar{1}$, implying that $|K : H| = 1$ which contradicts the assumption that $|K/H|$ is prime q . Thus, $|G : H| = p$ and so $|G : H|$ must be prime. ■

Problem 3. (a) Suppose G is a group with order p^3q where p and q are primes with $p > q$. Prove that if G has a subgroup H of order p^3 , then it must be normal.

Proof. Consider the quotient G/H . Then its order is

$$|G/H| = \frac{|G|}{|H|} = \frac{p^3q}{p^3} = q \implies G/H \text{ is cyclic.}$$

This tells us further that G/H is abelian, and so every subgroup of G/H is normal. By the fourth isomorphism theorem, we know that $H/H \trianglelefteq G/H$ implies that $H \trianglelefteq G$. ■

(b) Prove that if $H \leq Z(G)$, then G must be abelian.

Proof. Notice that the cosets of H in G partition G ; that is,

$$G = \bigcup_{g \in G} gH.$$

Now, let $x, y \in G$. Then for some $g_1, g_2 \in G$, we have $x \in g_1H$ and $y \in g_2H$. That is, for some $h_1, h_2 \in H$, we have $x = g_1h_1$ and $y = g_2h_2$. Now, we have since $h_1, h_2 \in H$ and $H \leq Z(G)$, we get that $h_1, h_2 \in Z(G)$ and so

$$\begin{aligned} xy &= (g_1h_1)(g_2h_2) \\ &= g_1(g_2h_1)h_2 \\ &= (h_1h_2)(g_1g_2) \\ &= (g_2h_2)(g_1h_1) \\ &= yx. \end{aligned}$$

Thus, $xy = yx$ and so G must be abelian. ■

Problem 4. Let G be a group and $g \in G$. Define $\sigma_g(x) = gxg^{-1}$ for all $x \in G$.

(a) Prove that the mapping $\psi : G \rightarrow S_G$ defined by $\psi(g) = \sigma_g$ is a homomorphism.

Proof. For all $g_1, g_2 \in G$ and for all $x \in G$, we have

$$\begin{aligned} \psi(g_1g_2)(x) &= \sigma_{g_1g_2}(x) = (g_1g_2)x(g_1g_2)^{-1} \\ &= g_1(g_2xg_2^{-1})g_1^{-1} \\ &= g_1(\sigma_{g_2}(x))g_1^{-1} \\ &= \sigma_{g_1}(\sigma_{g_2}(x)) \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(x) \\ &= (\psi(g_1) \circ \psi(g_2))(x). \end{aligned}$$

This implies that $\psi(g_1g_2) = \psi(g_1) \circ \psi(g_2)$ and so we conclude that ψ is a group homomorphism. ■

(b) Prove that $G/Z(G)$ is isomorphic to a subgroup of S_G .

Proof. Using part (a), it follows that

$$G/\ker(\psi) \cong \psi(G) \leq S_G.$$

From here, it suffices to show that $\ker(\psi) = Z(G)$. Indeed, we get that

$$\begin{aligned}\ker(\psi) &= \{g \in G : \psi(g)(x) = x \ \forall x \in G\} \\ &= \{g \in G : \sigma_g(x) = x \ \forall x \in G\} \\ &= \{g \in G : gxg^{-1} = x \ \forall x \in G\} \\ &= Z(G).\end{aligned}$$

Thus, we have

$$G/Z(G) \cong K \leq S_G$$

where $K = \psi(G)$. ■